

## SUMMARY TABLE: ESSENTIAL MATHEMATICS GRADE 10

### Sub-Strand 2.1: Similarity and Enlargement — Teacher Reference (Pre-filled)

| SUMMARY TABLE: ESSENTIAL MATHEMATICS GRADE 10 |                                                                                                                                                                                                         |
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| Sub-Strand                                    | Sub-Strand 2.1: Similarity and Enlargement                                                                                                                                                              |
| Driving Question                              | DRIVING QUESTION: Why does a sufuria that is twice as tall and twice as wide hold far more than twice the githeri, and how can we predict how length, area and volume change when we enlarge an object? |

### INSTRUCTIONS

FOR TEACHERS: This is the teacher reference version — each row is pre-filled with expected student responses. Use it to assess student Summary Tables, identify gaps, and prepare discussion questions. The student version has blank cells for students to complete after each lesson.

| Lesson # and Title                                                                                                | What did I observe?                                                                                                                                                                                                                                                                                                                                                                 | What did I learn?                                                                                                                                                                                                                                                                                           | How does this explain the phenomenon?                                                                                                                                                                                                                                                                                                                                                             |
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| <b>Lesson 1</b><br><b>Meet Mama Achieng's Three Sufurias: Which Feeds More, and What Does 'Similar' Mean?</b>     | We observed that the small, medium and large sufurias have the same shape: the height and diameter of the large one are both 2 times those of the small one (heights 12, 18, 24 cm; diameters 16, 24, 32 cm), while the price is KSh 60, 90 and 120. We also observed that some objects look the same shape in different sizes and others do not, even when both have right angles. | We learned that similar objects have the same shape: their corresponding angles are equal and their corresponding sides are in the same ratio. Equal angles alone are not enough, because a tall thin rectangle and a short wide rectangle are not similar.                                                 | So far we can explain that the three sufurias are similar because every length of the large one is 2 times the small one. We cannot yet explain why the large sufuria seems to feed far more than twice as many people, or why the paint costs more than double. Those questions are on our Driving Question Board and in our first model.                                                        |
| <b>Lesson 2</b><br><b>Properties of Similar Figures: Is the Ratio the Same Everywhere?</b>                        | We measured the sides and angles of rectangles A (4 cm by 6 cm), B (6 cm by 9 cm) and C (6 cm by 8 cm). The ratios were 1.5 and 1.5 for B to A, but 1.5 and about 1.33 for C to A. For the sufurias, height ratios (12, 18, 24 cm) and diameter ratios (16, 24, 32 cm) were both 1 : 1.5 : 2.                                                                                       | Similar figures have equal corresponding angles and all corresponding sides in the same ratio (a constant ratio). If even one pair of sides has a different ratio, the figures are not similar. The three sufurias are similar because every length of M is 1.5 times S and every length of L is 2 times S. | B is similar to A because both side ratios are 1.5. C is not similar to A because its side ratios (1.5 and about 1.33) are not the same, even though all three are rectangles. Mama Achieng's sufurias are similar because height and diameter both go up in the ratio 1 : 1.5 : 2. We still need to find out why this constant ratio of lengths does not make the githeri and paint just double. |
| <b>Lesson 3</b><br><b>Enlargement and Centre of Enlargement: Tracing Mama Achieng's Sufurias and Triangle ABC</b> | We traced triangle ABC (3, 4, 5 cm) and its image A'B'C' (6, 8, 10 cm), joined A to A', B to B' and C to C', and saw that the lines met at one point, O. All side ratios and OA'/OA, OB'/OB, OC'/OC were 2. The sufuria heights (12 to 24 cm) and diameters (16 to                                                                                                                  | An enlargement maps each point of an object to the image along a straight line from a fixed centre O. The centre is found by joining corresponding points. The linear scale factor $k = \text{image length} / \text{object length} = OA'/OA$ . For Mama Achieng's sufurias S                                | The large sufuria is an enlargement of the small one with $k = 2$ because every length (height and diameter) is multiplied by 2. This explains why the large sufuria looks like a doubled copy, but we still need to find out what $k = 2$ does to the paint area and                                                                                                                             |

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|                                                                                                                                                                      | 32 cm) also doubled.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | to L, $k = 2$ .                                                                                                                                                                                                                                                                                                                          | the capacity.                                                                                                                                                                                                                                                                                                                                                                                                          |
| <b>Lesson 4</b><br><b>Drawing the Image: Enlarging and Shrinking Triangle ABC from Centre O</b>                                                                      | We drew triangle ABC (3, 4, 5 cm) from centre O with $k = 2$ and got an image with sides 6, 8, 10 cm; with $k = 0.5$ we got sides 1.5, 2, 2.5 cm. On squared paper, rectangle A (4 cm by 6 cm) with $k = 2$ gave an image 8 cm by 12 cm. The angles stayed equal and the digital zoom matched our drawings.                                                                                                                                                                                                           | To draw the image, draw a ray from O through each object point, measure OA, calculate $OA' = k \times OA$ , mark A' on the ray and join the image points. When $k > 1$ the image is bigger and further from O; when $0 < k < 1$ it is smaller and nearer to O. Each image side is $k$ times the object side and the shape stays similar. | Mama Achieng's large sufuria is a $k = 2$ enlargement of the small one, so every length (height 12 to 24 cm, diameter 16 to 32 cm) is doubled, just like our triangle sides. This explains why it looks the same shape and why the KSh 120 price follows the doubled lengths. It does not yet explain why capacity and paint do not just double, which we will investigate with area (Lesson 5) and volume (Lesson 6). |
| <b>Lesson 5</b><br><b>Area Scale Factor: Why Paint Costs Rise Faster Than Price</b>                                                                                  | We enlarged a 3 cm by 2 cm rectangle (area 6 cm <sup>2</sup> ) with $k = 3$ to 9 cm by 6 cm (area 54 cm <sup>2</sup> ), so the area ratio was 9. We worked out the outside areas of the sufurias: S 803.84 cm <sup>2</sup> , M 1808.64 cm <sup>2</sup> , L 3215.36 cm <sup>2</sup> , with ratios to S of 1, 2.25 and 4, and paint costs of about KSh 20, KSh 45 and KSh 80.                                                                                                                                           | The area scale factor is image area divided by object area, and it equals $k^2$ . Doubling the lengths ( $k = 2$ ) multiplies the area by 4, and tripling ( $k = 3$ ) multiplies it by 9. Area depends on two lengths, so both are multiplied by $k$ . The price of KSh 120 follows only $k$ , but the paint follows $k^2$ .             | Mama Achieng's large sufuria is 2 times as tall and 2 times as wide as the small one, so its outside area is $2^2 = 4$ times larger. That is why painting it costs about KSh 80 against KSh 20, which is more than double. We still need to find out why the githeri capacity grows by even more (Lesson 6).                                                                                                           |
| <b>Lesson 6</b><br><b>Volume Scale Factor: Why the Large Sufuria Holds 8 Times the Githeri</b>                                                                       | We measured cubes (5 cm and 10 cm sides: 125 and 1000 cm <sup>3</sup> ) and cuboids (5 x 4 x 3 cm and 10 x 8 x 6 cm: 60 and 480 cm <sup>3</sup> ). Doubling every length ( $k = 2$ ) made the volume 8 times bigger. We also saw the sufuria capacities of about 2.4 litres, 8.1 litres and 19.3 litres and servings of about 6, 20 and 48.                                                                                                                                                                           | For similar solids the volume scale factor is $k^3$ . Area scale factor is $k^2$ (Lesson 5) and the linear scale factor is $k$ . Capacity depends on three lengths, so it multiplies by $k \times k \times k$ . The large sufuria ( $k = 2$ ) holds exactly 8 times the small one.                                                       | Mama Achieng's large sufuria is 2 times as tall and as wide, so its capacity is $2^3 = 8$ times the small one (about 48 servings against 6), not 2 times. The price (KSh 120) follows only $k = 2$ , so customers get a bargain (KSh 2.50 against KSh 10 per serving), while the paint area grows 4 times (KSh 80 against KSh 20).                                                                                     |
| <b>Lesson 7</b><br><b>Which Factor Do I Use? Moving Between <math>k</math>, <math>k^2</math> and <math>k^3</math> with Posters, Maps and Mama Achieng's Sufurias</b> | We observed that when the photo is enlarged with $k = 3$ the area multiplies by 9 (150 cm <sup>2</sup> to 1350 cm <sup>2</sup> ); on the 1 : 50 000 map 1 cm is 500 m but 1 cm <sup>2</sup> is 250 000 m <sup>2</sup> (25 hectares); and for Mama Achieng's sufurias the large one ( $k = 2$ ) costs only 2 times as much (KSh 120 vs KSh 60) but holds 8 times as much (about 48 servings against 6) and needs about 4 times the paint (about KSh 80 against KSh 20), so a serving costs KSh 2.50 instead of KSh 10. | We learned that a length uses $k$ , an area uses $k^2$ and a volume uses $k^3$ , and that we can work backwards: an area factor of 9 means $k = 3$ , and a volume factor of 27 means $k = 3$ . We learned to ask first whether a quantity is a length, an area or a volume.                                                              | We can explain that the large sufuria holds far more than twice the githeri because capacity depends on three lengths (radius, radius and height), so it multiplies by $k^3 = 8$ , while paint depends on two lengths so it multiplies by $k^2 = 4$ , and the price only follows $k = 2$ . This is why large customers get a bargain while Mama Achieng's paint costs rise faster than her price.                      |
| <b>Lesson 8</b><br><b>Project: Make Similar Containers and Give the Final Explanation of the Sufuria Phenomenon</b>                                                  | We built a small and a larger container with all lengths in the same ratio (for example cylinder tins with radius 4 cm and height 10 cm, and radius 6 cm and height 15 cm, $k = 1.5$ ; or cubes of side 5 cm and 10 cm). We                                                                                                                                                                                                                                                                                           | When an object is enlarged by a linear scale factor $k$ , lengths multiply by $k$ , areas multiply by $k^2$ and volumes multiply by $k^3$ . Our cylinders: $k = 1.5$ , volume 502.4 cm <sup>3</sup> to 1695.6 cm <sup>3</sup> , ratio $3.375 = 1.5^3$ .                                                                                  | The large sufuria is similar to the small one with $k = 2$ . Its capacity depends on three lengths (radius, radius, height), so it holds exactly 8 times as much: 19292.16 cm <sup>3</sup> against 2411.52 cm <sup>3</sup> , about 48 servings                                                                                                                                                                         |

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|  | <p>measured them, filled the small one with maize or sand and poured it into the larger one, and recorded the process with photos or video. We also reviewed Mama Achieng's three sufurias: S, M and L, with L twice as tall and twice as wide as S.</p> | <p>Our cubes: <math>k = 2</math>, volume 125 cm<sup>3</sup> to 1000 cm<sup>3</sup>, ratio 8. A quantity depends on as many lengths as it has dimensions: one length, two lengths or three lengths.</p> | <p>against 6. Its outside area depends on two lengths, so it is 4 times as large (3215.36 cm<sup>2</sup> against 803.84 cm<sup>2</sup>), and paint costs about KSh 80 against KSh 20. The price depends only on the linear factor (KSh 60 x 2 = KSh 120), so customers get a bargain (KSh 2.50 per serving against KSh 10) while Mama Achieng's paint cost rises faster than her price.</p> |
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## ATTRIBUTION AND LICENSING

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